

PROJECT ZEPHYRAQUUS

ENGINEERING NOTEBOOK

PROTOTYPE V0.3 – ESC INTEGRATION & WIRING ARCHITECTURE

Objective-003:

Determine the final ESC locations, power distribution layout, signal routing, and wiring plan prior to electrical assembly.

DG-003:

- Position both SpeedyBee F405 stacks.
- Minimize motor wire lengths.
- Maintain center of gravity.
- Keep wiring organized and serviceable.
- Verify no wire interference with propellers.

DD-003:

1. The ESCs were positioned to align with the inward-facing motor leads, minimizing motor wire length, shown in the left photo in Fig. 3.
2. The battery will be oriented horizontally and positioned at the center of gravity to distribute power and mass evenly between both ESCs.
3. This layout minimizes wire length, reduces crossing wires, and keeps the electrical system easier to inspect and maintain.

ECE-003:

- XT60 connector gender is only relevant at mating connectors; the ESC power pads accept soldered wires directly..

Figure 3. Prototype layout and preliminary wiring architecture prior to electrical assembly.

